

Department Of Computer Science

MCA - III

Subject: Enterprise Java Technologies

JSP Theory Assignment – 4

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1. Explain JSP and discuss its advantages over traditional servlet-based web development.
2. Explain the architecture and life cycle of a JSP page with a suitable diagram.
3. Explain the different types of JSP scripting elements with suitable examples.
4. Explain JSP directives. Discuss the different types of directives with examples.
5. Explain the JSP page directive and describe its commonly used attributes.
6. Explain the JSP include directive and `<jsp:include>` action. Differentiate between them.
7. Explain JSP standard actions. Discuss `<jsp:useBean>`, `<jsp:setProperty>` and `<jsp:getProperty>`.
8. Explain the concept of JavaBeans in JSP. Discuss the advantages of using JavaBeans with JSP.
9. Explain JSP implicit objects. Describe any five implicit objects with their uses.
10. Explain the `request`, `response`, `session`, `application` and `out` implicit objects in detail.
11. Explain different techniques for session tracking in JSP.
12. Explain cookies and `HttpSession` in JSP. Compare their usage and characteristics.
13. Explain JSP exception handling. Discuss the use of `errorPage` and `isErrorPage` attributes.
14. Explain the role of JSP in MVC architecture. Describe how JSP, Servlet and JavaBean can work together.
15. Explain the complete translation and execution process of a JSP page by the JSP container.
16. What is Expression Language (EL)? Explain its features and advantages in JSP.
17. Explain the syntax and basic rules of JSP Expression Language with suitable examples.
18. Explain the different scopes available in EL:
 - Page Scope
 - Request Scope
 - Session Scope
 - Application Scope
19. Explain EL implicit objects. Discuss objects such as `param`, `paramValues`, `header`, `cookie`, `initParam` and `pageContext`.
20. Explain arithmetic, relational, logical and conditional operators available in Expression Language.
21. Explain how EL is used to access JavaBean properties with suitable examples.
22. Explain how EL can be used to access arrays, lists, maps and object properties.
23. Differentiate between JSP scriptlets and Expression Language. Explain why EL is preferred in modern JSP applications.
24. What is JSTL? Explain its purpose, features and advantages in JSP development.
25. Explain the different tag libraries provided by JSTL.
26. Explain the JSTL Core tag library and discuss the following tags:
 - `<c:set>`
 - `<c:out>`
 - `<c:remove>`
 - `<c:if>`
27. Explain `<c:choose>`, `<c:when>` and `<c:otherwise>` with suitable examples.
28. Explain the `<c:forEach>` tag in JSTL. Discuss its important attributes and applications.
29. Explain the use of JSTL `<c:forTokens>` with a suitable example.
30. Explain JSTL URL-related tags. Discuss `<c:url>`, `<c:redirect>` and `<c:param>`.
31. Explain how JSTL and EL can be used together to develop scriptlet-free JSP pages.
32. Differentiate between JSP Scriptlet, EL and JSTL with respect to syntax, purpose, readability and maintainability.
33. Explain how JSP, JSTL and EL can be integrated with JDBC to display database records dynamically.
34. Explain the advantages and limitations of using JSP, JSTL and EL in enterprise web applications.
35. Explain the complete request-processing flow of a web application using `HTML → Servlet → JDBC → JSP → JSTL → EL`, with a suitable architecture diagram.